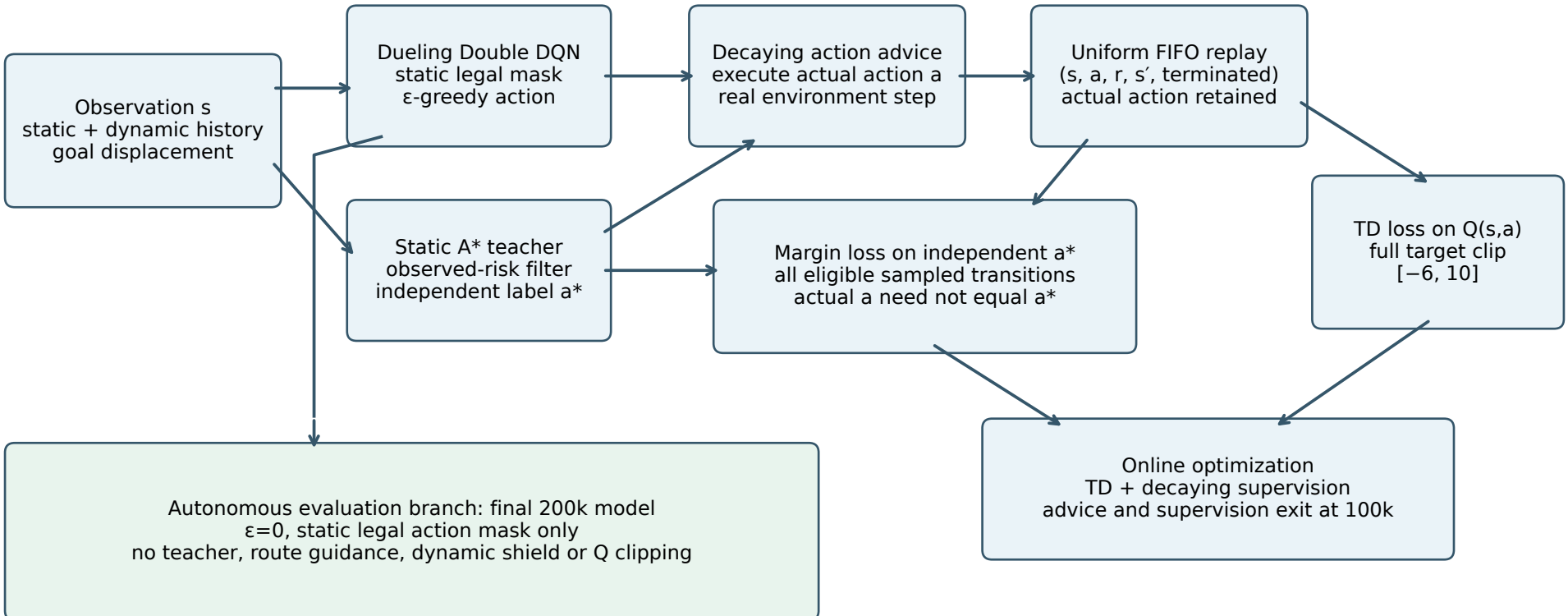
Observation s
static + dynamic history
goal displacement
Dueling Double DQN
static legal mask
ε-greedy action
Decaying action advice
execute actual action a
real environment step
Uniform FIFO replay
(s, a, r, s′, terminated)
actual action retained
Static A* teacher
observed-risk filter
independent label a*
Margin loss on independent a*
all eligible sampled transitions
actual a need not equal a*
TD loss on Q(s,a)
full target clip
[−6, 10]
Autonomous evaluation branch: final 200k model
ε=0, static legal action mask only
no teacher, route guidance, dynamic shield or Q clipping
Online optimization
TD + decaying supervision
advice and supervision exit at 100k